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Seed inoculation with endophytic fungal entomopathogens promotes plant growth and reduces crown and root rot (CRR) caused by *Fusarium culmorum* in wheat

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Abstract

Main conclusion Fungal entomopathogens, *Beauveria bassiana* (NATURALIS) and *Metarhizium brunneum* (BIPECO5), can promote the growth of wheat following their endophytic establishment within plants through seed treatment. Similar to endophytic *B. bassiana* which has already been reported as a disease antagonist by several previous studies, the present study demonstrates that *M. brunneum* can suppress disease pathogens following plant colonization as well.

An upsurge of research hints at the ability of entomopathogenic fungi, almost exclusively considered and used as insect pathogens, to endophytically colonize the internal tissues of a wide array of host plants and subsequently confer numerous benefits including enhancement of plant growth and suppression of disease pathogens. Such an ability has mainly been investigated for *Beauveria bassiana*. Fewer studies have demonstrated plant growth promotion by *Metarhizium brunneum* colonization, whereas no studies have reported on the potential of endophytic *M. brunneum* as a plant disease antagonist. The present study was, therefore, conducted to investigate whether seed treatment with *B. bassiana* (NATURALIS) and *M. brunneum* (BIPECO5) could result in their endophytic establishment in wheat and promote plant growth. The study further examines the effect of the fungal strains as endophytes against *Fusarium culmorum*, one of the main causal agents of crown and root rot (CRR) in wheat. Both *B. bassiana* and *M. brunneum* were able to systemically colonize roots and shoots of wheat, and promote several plant growth parameters (shoot height, root length, and fresh root and shoot weights). Moreover, endophytic colonization of wheat with either fungal entomopathogen resulted in a significant reduction in disease incidence, development and severity. These results support the notion of the multiple ecological roles that could further be played by entomopathogenic fungi. Bearing such additional roles in mind while developing these fungi as microbial agents could improve the value of many commercially available mycoinsecticides.

Keywords *Beauveria bassiana* · Entomopathogenic fungi as endophytes · *Fusarium* crown rot of wheat · *Metarhizium brunneum* · Plant disease biocontrol · Plant growth promotion

Abbreviations

IPM	Integrated pest management
CRR	Crown and root rot
FRR	Foot and root rot
FCR	<i>Fusarium</i> crown rot
FHB	<i>Fusarium</i> head blight
PDA	Potato dextrose agar

CRD	Completely randomized design
DPI	Days post-inoculation
SAS	Statistical analysis system
ZYMV	Zucchini yellow mosaic virus
FCRR	<i>Fusarium</i> crown and root rot

Introduction

Studies have revealed that entomopathogenic fungi, typically known as important natural enemies of arthropods and mainly developed as biopesticides against insect pests (Lacey et al. 2015), are able to play additional roles

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as endophytes, plant growth promoters, disease antagonists, and rhizosphere colonizers (Vega et al. 2009). Such diverse ecological functions recently discovered for fungal entomopathogens suggest that they have the potential to be developed as versatile microbial agents with multiple uses in sustainable agriculture, for example as dual-biocontrol agents of insect and pathogen pests as well as biofertilizers (Vega et al. 2009; Lacey et al. 2015; Jaber and Ownley 2018). There has been an upsurge of research on the ability of various genera of entomopathogenic fungi to endophytically colonize the internal tissues of a wide array of host plants (Vega 2008; Gurulingappa et al. 2010; Akutse et al. 2013) and subsequently confer numerous benefits including enhancement of plant growth (Lopez and Sword 2015; Jaber and Enkerli 2016, 2017) and suppression of disease pathogens (Ownley et al. 2008, 2010; Jaber and Ownley 2018), in addition to their more widely known role in arthropod pest control following conventional inundative releases (de Faria and Wraight 2007). Also, the use of entomopathogenic fungi as endophytes has been found compatible with other integrated pest management (IPM) tools such as parasitoids and botanicals (Akutse et al. 2014; Gathage et al. 2016; Jaber and Araj 2018). Therefore, a deeper understanding of the various ecological roles or potential interactions these fungi might display post-application would be important for the development of more effective control approaches and sustainable production practices (Vega et al. 2009; Lacey et al. 2015; Jaber and Ownley 2018).

Enhanced plant growth, mediated by the endophytic colonization with different genera of fungal entomopathogens, has been demonstrated following fungal inoculation via various methods such as seed treatment, foliar spray, and root drench (Gurulingappa et al. 2010; Garcia et al. 2011; Lopez and Sword 2015; Jaber and Enkerli 2016, 2017; Jaber and Araj 2018). For instance, seed inoculation of cotton with *Beauveria bassiana* and *Purpureocillium lilacinum* (formerly *Paecilomyces lilacinus*) had significant growth-promoting effects on plant dry biomass, number of nodes, and number of reproductive tissues (Lopez and Sword 2015). Similarly, treatment of broad bean seeds with *B. bassiana* and *Metarhizium brunneum* significantly enhanced plant height, leaf pair number, and fresh shoot and root weights (Jaber and Enkerli 2016). Also following a pre-treatment with fungal inoculum, increased resistance to fungal, bacterial, and viral plant pathogens has been reported for plants endophytically colonized with fungal entomopathogens in several genera (Jaber and Ownley 2018). Of these, *B. bassiana* is the most extensively studied entomopathogen as a plant disease antagonist. Endophytic *B. bassiana* has been reported to suppress damping off caused by the soil-borne pathogens *Rhizoctonia solani* and *Pythium myriotylum* in tomato (Ownley et al. 2004) and cotton (Ownley et al. 2008), bacterial blight caused by *Xanthomonas axonopodis* pv.

malvacearum in cotton (Ownley et al. 2008), the Zucchini yellow mosaic virus in squash (Jaber and Salem 2014), and downy mildew caused by *Plasmopara viticola* in grapevines (Jaber 2015). In addition to *B. bassiana*, antagonistic activity against plant disease pathogens has also been reported for endophytic *Metarhizium robertsii* against the root rot pathogen *Fusarium solani* f. sp. *phaseoli* in bean (Sasan and Bidochka 2013), and endophytic *Lecanicillium* spp. against *Pythium ultimum* (Benhamou and Brodeur 2001) and powdery mildew *Podosphaera fuliginea* (previously known as *Sphaerotheca fuliginea*) in cucumber (Hirano et al. 2008), but has so far never been documented for endophytic *M. brunneum*. Also notable is the fact that, although the ability of fungal entomopathogens applied as endophytes to act as plant disease antagonists has been demonstrated against several plant pathogens, it has not yet been shown against *Fusarium culmorum*, one of the main causal agents of crown and root rot (CRR) of cereals.

CRR, also known as foot and root rot (FRR) or Fusarium crown rot (FCR), is an important disease complex attacking different small-grain cereals, particularly wheat and barley (Scherin et al. 2013). It is caused by the ubiquitous soil-borne fungus *F. culmorum*, which has been considered as one of the main pathogens of wheat worldwide (Burgess et al. 2001; Wagacha and Muthomi 2007; Scherin et al. 2013). The pathogen infects wheat plants at an early growth stage and causes rotting of root, crown, and lower stem tissues, resulting in stand reductions and yield losses. However, yield and quality losses become particularly significant when the pathogen infects plants at the flowering stage (i.e., anthesis), causing Fusarium head blight (FHB), which is also known as ear blight or scab (Scherin et al. 2013). The latter disease results in grain contamination with mycotoxins (Wagacha and Muthomi 2007). Due to potential health hazards of mycotoxin contamination of grains for both humans and animals, control of FHB in cereals has received more research attention than CRR control (Khan et al. 2006). Potential options for *F. culmorum* control include the use of fungicides, resistant cultivars, cultural practices, and biological control agents (Wagacha and Muthomi 2007; Scherin et al. 2013). Because of the risks associated with fungicide toxic residues and resistance development, in addition to the absence of highly resistant commercial wheat cultivars, chemical seed treatment and cultivar resistance have only shown limited success (Khan et al. 2006; Wagacha and Muthomi 2007; Scherin et al. 2013). Disease management has thus focused on cultural control practices, which are often related to moisture management (Cook 1980). The integration of biological control agents has been found effective in supporting other control measures used for *F. culmorum* management (Wagacha and Muthomi 2007; Scherin et al. 2013). Pre-treatment of different plant organs with biocontrol agents via spray, drench, or seed coating

reduced disease severity on wheat and other cereals (e.g., Kempf and Wolf 1989; Luongo et al. 2005; Khan et al. 2006; Serfling et al. 2007; Khan and Doohan 2009; Scherm et al. 2013). In particular, several previous studies, reviewed in Scherm et al. (2013), have shown that seed treatment with bacterial (e.g., *Bacillus* spp. and fluorescent pseudomonads) and fungal (e.g., *Trichoderma* spp. and *Clonostachys rosea*) microbial agents is promising for the biological control of *F. culmorum* in wheat. Therefore, the present study was conducted to investigate whether seed treatment of wheat with fungal entomopathogens (*B. bassiana* NATURALIS and *M. brunneum* BIPESCO5) could (1) result in their endophytic establishment in wheat, (2) promote plant growth, and (3) reduce CRR disease symptoms caused by *F. culmorum* infection. These objectives were addressed under controlled greenhouse conditions. The two tested entomopathogenic fungal strains have been shown to endophytically colonize different host plants and subsequently confer several benefits such as improved plant growth and enhanced protection against insect pests and plant diseases (e.g., Jaber and Salem 2014; Jaber 2015; Jaber and Enkerli 2016, 2017; Jaber and Araj 2018; Jaber et al. 2018). However, neither *B. bassiana* NATURALIS nor *M. brunneum* BIPESCO5 have been previously reported as endophytes in wheat.

Materials and methods

Plant and fungal material

Seeds of wheat (*Triticum aestivum* L.) cv. Norsì (National Center for Agricultural Research and Extension, Jordan), which is highly susceptible to most soil-borne fungal pathogens, were used for all experiments. Prior to use, seeds were surface sterilized as described in Fernandez and Chen (2005) with minor modifications. Briefly, seeds were soaked for 2 min in an aqueous solution of 0.6% sodium hypochlorite (NaOCl), then for 2 min in 70% ethanol, and finally rinsed thrice in sterile distilled water. Samples taken from the third rinse water were plated onto potato dextrose agar (PDA) medium to confirm the efficiency of the surface sterilization procedure (Schulz et al. 1998). *B. bassiana* strain ATCC 74040, registered as NATURALIS® (Intrachem Bio Italia S.p.A., Bergamo, Italy), and *M. brunneum* strain BIPESCO5 (CBS123710), registered for pest biocontrol in Europe, were used. Both fungal strains were cultured and maintained on PDA plates at 25 °C in complete darkness. An isolate of *F. culmorum*, originally obtained from severely rotted crown and root tissues of wheat and previously identified as described in Jaber and Alananbeh (2018), was included in the present study. Fungal cultures of *F. culmorum* grown on PDA plates for about 2 weeks at 25 °C were used for preparation of the fungal disease inoculum as described below.

Seed inoculation with fungal entomopathogens and assessment of endophytic colonization and plant growth

Seed inoculation was carried out as described in Jaber and Enkerli (2016). Fungal conidia were harvested under sterile conditions by gently scraping the surface of 14-day-old sporulating cultures of *B. bassiana* and *M. brunneum* after flooding the plates with sterile distilled water containing Tween 80 (0.1% v/v, Difco™). The resulting mycelia and spores were then filtered through several layers of sterile cheese cloth into a sterile glass bottle containing 250 ml sterile distilled water plus 0.1% Tween 80. The conidial suspension of each fungal strain was then homogenized with a magnetic stirrer for 5 min. Conidial concentration for each fungus was determined using a Neubauer hemocytometer and adjusted to a concentration of 1×10^7 conidia ml⁻¹. This inoculum concentration was successfully used for the endophytic establishment of different fungal entomopathogens, including the fungal strains tested here, in several previous studies (e.g., Gurulingappa et al. 2010; Castillo-Lopez et al. 2014; Lopez and Sword 2015; Jaber and Enkerli 2016, 2017; Jaber and Araj 2018; Jaber et al. 2018). Conidial viability for each fungal strain was determined before bioassay by carrying out a germination test. A low concentration of conidia was suspended in 0.1% Tween 80 solution, and a 100 µl aliquot from this stock was plated onto PDA and incubated at 25 °C for 24 h. Three randomly selected groups of 100 conidia were assessed for germination. Conidia were deemed to have germinated only when the length of the germ tube was at least twice the diameter of the conidium. An average of three replicate counts was calculated for each fungal strain. Germination exceeded 90% for each strain on average. For plant inoculation, surface-sterilized seeds were soaked in the conidial suspension of each fungal strain or in sterile distilled water containing 0.1% Tween 80 (control). Bottles containing the soaking seeds were kept for 16 h in the dark at 25 °C. Seeds were then sowed about 2 cm deep in potted planting substrate (a mixture of soil, sand, and peat at a ratio of 1:1:1). Prior to potting, the planting substrate was sterilized by autoclaving at 121 °C for 2 h and allowed to cool for 24 h. Four seeds were planted per pot. Pots were maintained under controlled conditions in the greenhouse (22 ± 3 °C, 40–50% relative humidity, and a photoperiod of 16:8 h light/dark). Plants were irrigated with sterile distilled water as needed, but never fertilized during the course of experiments. Experiments were run using a Completely Randomized Design (CRD) with six treatments: (1) NATURALIS + *F. culmorum*, (2) BIPESCO5 + *F. culmorum*, (3) *F. culmorum*, (4) NATURALIS, (5) BIPESCO5, and (6) CONTROL. Each treatment had a total of 20 seeds ($n = 20$). All experiments were independently conducted twice over time (Experiment 1 and Experiment 2).

The endophytic colonization of plants by *B. bassiana* and *M. brunneum* was determined by destructively sampling root and shoot tissues of plants in the following treatments: NATURALIS, BIPESCO5, and CONTROL. Half of the plants per each of these treatments was sampled by the time of *F. culmorum* infection (i.e., 14 days post-inoculation; 14 dpi), whereas the other half was sampled at the end of experiments (i.e. 24 days post-inoculation; 24 dpi) as described below. Several plant growth parameters (shoot height, root length, and fresh root and shoot weights) were measured before destructively sampling plants for assessment of their endophytic colonization by the inoculated entomopathogenic fungal strains. Shoots and roots (after thorough washing with running tap water) were surface sterilized by submerging for 2 min in 0.5% sodium hypochlorite for shoot tissues and in 1% sodium hypochlorite for root tissues, followed by 2 min in 70% ethanol and three rinses in sterile distilled water. Plant material was then allowed to surface-dry in a laminar flow hood. An average of ten shoot and root tissue pieces (approximately 8–10 mm long) were cut from each surface-sterilized plant tissue using a sterile scalpel. Sampled plant tissues were then evenly plated on 90-mm Petri dishes containing semi-selective culture medium (Strasser et al. 1996) that was slightly modified according to Posadas et al. (2012). The efficiency of the surface sterilization procedure was assessed by making imprints of surface-sterilized plant tissue and plating the final rinse water onto PDA (Schulz et al. 1998). Plates were incubated at 25 °C and inspected every 2–3 days for 3 weeks to observe and record fungal outgrowth. Fungal outgrowth from the plated plant samples were identified as *B. bassiana* or *M. brunneum* based on differential growth on the semi-selective media, colony morphology, and microscopic examination of conidia (Humber 1997). Percent colonization of plant tissues by the respective inoculated fungus was calculated as number of sampled plant tissue showing fungal outgrowth divided by the total number of plated plant tissue samples $\times$ 100 (Petrini and Fisher 1987). Presence of other bacterial or fungal endophytes on the incubated plates was occasionally observed, but not scored.

***F. culmorum* infection and CRR disease assessment**

14 days post-inoculation (14 dpi) of wheat seeds with *B. bassiana* and *M. brunneum*, when wheat seedlings had two unfolded leaves (Zadoks' GS 12; Zadoks et al. 1974), each seedling was infected with 3 ml of *F. culmorum* suspension (2×10^5 macroconidia ml⁻¹) as described in Motallebi et al. (2015). Briefly, the planting substrate around the seedling was removed and the inoculum was applied with a pipette along the stem, 1–2 cm above the planting substrate line, to allow the *F. culmorum* suspension to reach the stem bases. The planting substrate was then replaced around the plant

stem. Disease incidence was non-destructively monitored on a daily basis. Days to symptom development and the proportion of symptomless plants were recorded. 10 days following *F. culmorum* infection, when almost all *F. culmorum*-infected plants (positive control) developed severe symptoms and some were completely rotted or dead, plants in all treatments were carefully removed from the planting substrate and washed. Disease severity was assessed using an index from 0 to 4, as previously described in several studies (e.g., Fernandez and Chen 2005; Khan et al. 2006; Motallebi et al. 2015): 0 = healthy plants, 1 = mild (< 25%) browning on stem and/or root, 2 = moderate (25–50%) browning on stem and/or root, 3 = severe (> 50%) browning on stem and/or root, and 4 = same as 3 but plant tissue was completely rotted or dead. Plant growth parameters (plant height, root length, and fresh shoot and root weights) were also determined at the end of experiments. Finally, re-isolation of *F. culmorum* from infected crown and root tissues was done as described in Motallebi et al. (2015).

Statistics

All experiments were conducted twice (Experiment 1 and Experiment 2). Prior to statistical analyses, all data were tested for normality and homogeneity of the variance. Percentage values of plant tissue colonization with *B. bassiana* and *M. brunneum* in addition to CRR incidence were positively skewed (Skewness > 0) and showed non-Gaussian distributions (Kolmogorow–Smirnow, $P < 0.05$). Therefore, transformation with a natural logarithm (ln) function [$\ln(y) = \text{Log}(y + 10)$] was applied to correct for heterogeneity of the variance and produce an approximately normally distributed data sets. Data from both experiments were subjected to one-way ANOVA. When a significant F test was obtained at $P = 0.05$, separation of treatment means was performed using Duncan's multiple range test. The Statistical Analysis System (SAS) software Version 9.1.3 (SAS Institute Inc. 2010) was used for conducting all statistical analyses.

Results

Endophytic colonization of wheat with fungal entomopathogens and plant growth promotion

Endophytic colonization of wheat plants with the two inoculated strains of entomopathogenic fungi was confirmed at 14 (i.e., the time of *F. culmorum* infection) and 24 (i.e., the end of all experiments) dpi by re-isolation of respective fungal strains on selective culture medium. *B. bassiana* (NATURALIS) and *M. brunneum* (BIPESCO5) were successfully recovered from all inoculated plants on

both sampling dates (Table 1). On the other hand, neither of the two strains was ever isolated from control plants. No fungal growth was observed on any of the imprint plates or on plates inoculated with final rinse water, demonstrating the efficiency of the surface sterilization procedure and confirming that the fungi growing out of surface-sterilized plant material were endophytic microorganisms that originated from within plant tissues. Also, re-isolation of both fungal strains from shoots and roots of seed-inoculated plants indicates a systemic plant colonization with these fungal entomopathogens as endophytes (Table 1). However, both *B. bassiana* (NATURALIS) and *M. brunneum* (BIPESCO5) significantly ($P < 0.001$) colonized more root than shoot tissues on both sampling dates. Despite a general decline in shoot and root colonization over time (Table 1), no significant difference in percentage plant colonization with either fungal strain was observed across both sampling dates (14 and 24 dpi).

Several growth parameters of wheat were significantly ($P < 0.05$) enhanced by the endophytic colonization of plants with the tested fungal entomopathogens (Fig. 1; Tables 2, 3). Inoculation of wheat seeds with either *B. bassiana* (NATURALIS) or *M. brunneum* (BIPESCO5) increased percentage seedling emergence, but such an increase was significant ($P = 0.01$) only in Experiment 2 (Fig. 1a). Seed inoculation with the tested strains also resulted in a significant decrease in days to seedling emergence in Experiments 1 and 2, irrespective of the strain being inoculated ($P < 0.01$; Fig. 1b). At 14 dpi (i.e., the time plants were infected with *F. culmorum*), wheat plants colonized with either *B. bassiana* (NATURALIS) or *M. brunneum* (BIPESCO5) had significantly higher shoot height, root length, and fresh shoot and root weights compared to non-colonized (control) plants (Table 2). Similarly at 24 dpi (i.e., after *F. culmorum* infection), all measured plant parameters were significantly enhanced in plants colonized with either entomopathogenic fungal strain compared to non-colonized controls (Table 3);

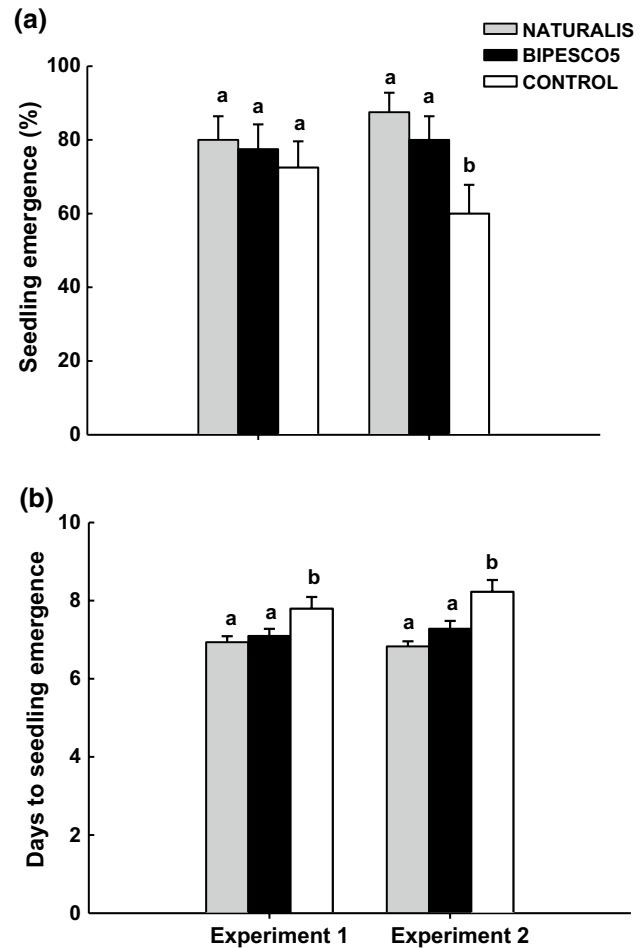


Fig. 1 Effect of wheat seed treatment with fungal entomopathogens (*B. bassiana* NATURALIS and *M. brunneum* BIPESCO5) on **a** mean ($\pm$ SE) percent seedling emergence and **b** mean ($\pm$ SE) days to seedling emergence. Bars with different letters within the same experiment differ significantly at $P = 0.05$ (Duncan's multiple range test after one-way ANOVA)

Table 1 Mean ($\pm$ SE) percent colonization of shoot and root tissues of wheat (*Triticum aestivum*) 14 and 24 days post-inoculation (dpi) with fungal entomopathogens (*Beauveria bassiana* strain NATURALIS and *Metarhizium brunneum* strain BIPESCO5)

Experiment	Plant tissue	<i>Beauveria bassiana</i> (NATURALIS)		<i>Metarhizium brunneum</i> (BIPESCO5)		Control	
		14 dpi	24 dpi	14 dpi	24 dpi	14 dpi	24 dpi
1	Shoot	33.1 $\pm$ 11a	28.6 $\pm$ 7a	24.3 $\pm$ 8a	20.3 $\pm$ 9a	0 $\pm$ 0b	0 $\pm$ 0b
	Root	61.3 $\pm$ 9a	57.5 $\pm$ 14a	50.7 $\pm$ 10a	45.9 $\pm$ 13a	0 $\pm$ 0b	0 $\pm$ 0b
2	Shoot	41.9 $\pm$ 10a	33.5 $\pm$ 8a	36.8 $\pm$ 8a	22.9 $\pm$ 7a	0 $\pm$ 0b	0 $\pm$ 0a
	Root	79.4 $\pm$ 8a	62.2 $\pm$ 12a	69.2 $\pm$ 9a	51.2 $\pm$ 12a	0 $\pm$ 0b	0 $\pm$ 0b

Endophytic plant colonization with the inoculated fungal strains was determined at the time wheat plants were infected with *Fusarium culmorum* (14 dpi) and at the end of all experiments (24 dpi). Experiments were conducted twice (Experiments 1 and 2 are presented)

For each plant tissue, means ($\pm$ SE) followed by the same letter within the same sampling date are not significantly different at $P = 0.05$ (Duncan's multiple range test after one-way ANOVA)

Table 2 Mean ($\pm$ SE) growth parameters of wheat sampled 14 days post-inoculation (14 dpi) with fungal entomopathogens (*B. bassiana* NATURALIS and *M. brunneum* BIPESCO5) and prior to infection with *F. culmorum*, the causal agent of crown and root rot (CRR)

Plant growth parameter	Experiment 1			Experiment 2		
	<i>B. bassiana</i> (NATURALIS)	<i>M. brunneum</i> (BIPESCO5)	Control	<i>B. bassiana</i> (NATURALIS)	<i>M. brunneum</i> (BIPESCO5)	Control
Shoot height (cm)	25.6 $\pm$ 1.2a	25.1 $\pm$ 1.0a	14.6 $\pm$ 0.7b	26.2 $\pm$ 1.2a	25.7 $\pm$ 1.0a	14.1 $\pm$ 1.5b
Root length (cm)	13.5 $\pm$ 1.1a	12.8 $\pm$ 1.3a	7.6 $\pm$ 0.6b	14.0 $\pm$ 1.4a	13.3 $\pm$ 1.2a	7.4 $\pm$ 0.7b
Fresh shoot weight (g)	0.21 $\pm$ 0.03a	0.22 $\pm$ 0.03a	0.14 $\pm$ 0.01b	0.22 $\pm$ 0.03a	0.21 $\pm$ 0.03a	0.13 $\pm$ 0.01b
Fresh root weight (g)	0.18 $\pm$ 0.02a	0.16 $\pm$ 0.02a	0.09 $\pm$ 0.01b	0.19 $\pm$ 0.03a	0.19 $\pm$ 0.01a	0.08 $\pm$ 0.01b

Experiments were conducted twice (Experiments 1 and 2 are presented)

For each plant growth parameter, means ($\pm$ SE) followed by the same letter within the same experiment are not significantly different at $P=0.05$ (Duncan's multiple range test after one-way ANOVA)

Table 3 Mean ($\pm$ SE) growth parameters of wheat sampled 24 days post-inoculation (dpi) with fungal entomopathogens (*B. bassiana* NATURALIS and *M. brunneum* BIPESCO5) and 10 days following infection with *F. culmorum*, the causal agent of crown and root rot (CRR)

Experiment	Treatment	Plant growth parameter			
		Shoot height (cm)	Root length (cm)	Fresh shoot weight (g)	Fresh root weight (g)
1	<i>F. culmorum</i> + NATURALIS	26.9 $\pm$ 1.1ab	15.8 $\pm$ 1.4ab	0.25 $\pm$ 0.03a	0.18 $\pm$ 0.02ab
	<i>F. culmorum</i> + BIPESCO5	28.3 $\pm$ 1.7a	15.1 $\pm$ 1.1ab	0.26 $\pm$ 0.03a	0.21 $\pm$ 0.06a
	<i>F. culmorum</i>	17.8 $\pm$ 2.4c	8.3 $\pm$ 0.7c	0.09 $\pm$ 0.02b	0.07 $\pm$ 0.01c
	NATURALIS	31.9 $\pm$ 2.1a	18.3 $\pm$ 1.1a	0.29 $\pm$ 0.03a	0.23 $\pm$ 0.04a
	BIPESCO5	31.3 $\pm$ 2.3a	17.9 $\pm$ 1.4a	0.30 $\pm$ 0.03a	0.26 $\pm$ 0.04a
	CONTROL	21.8 $\pm$ 1.5bc	12.5 $\pm$ 1.3b	0.20 $\pm$ 0.04a	0.15 $\pm$ 0.02b
2	<i>F. culmorum</i> + NATURALIS	27.2 $\pm$ 1.4ab	16.2 $\pm$ 1.3a	0.25 $\pm$ 0.04a	0.25 $\pm$ 0.03a
	<i>F. culmorum</i> + BIPESCO5	27.9 $\pm$ 1.8ab	16.1 $\pm$ 1.5a	0.24 $\pm$ 0.03a	0.23 $\pm$ 0.04a
	<i>F. culmorum</i>	17.6 $\pm$ 2.8c	8.1 $\pm$ 0.4b	0.08 $\pm$ 0.02c	0.06 $\pm$ 0.01c
	NATURALIS	32.3 $\pm$ 2.0a	18.6 $\pm$ 0.9a	0.30 $\pm$ 0.05a	0.28 $\pm$ 0.05a
	BIPESCO5	31.6 $\pm$ 2.7a	16.9 $\pm$ 1.0a	0.30 $\pm$ 0.05a	0.26 $\pm$ 0.04a
	CONTROL	21.4 $\pm$ 1.9bc	11.0 $\pm$ 1.2b	0.18 $\pm$ 0.04b	0.13 $\pm$ 0.02b

Experiments were conducted twice (Experiments 1 and 2 are presented)

In a column, means ($\pm$ SE) followed by the same letter are not significantly different at $P=0.05$ (Duncan's multiple range test after one-way ANOVA)

regardless of the presence or absence of *F. culmorum* infection (see below).

Effect of endophytic colonization of wheat with fungal entomopathogens on CRR caused by *F. culmorum*

Following *F. culmorum* infection, symptoms related to CRR were significantly ($P < 0.01$) lower on plants colonized with the inoculated fungal entomopathogens compared to non-colonized plants (control), irrespective of fungal strain (Fig. 2). Endophytic colonization of wheat with either *B. bassiana* (NATURALIS) or *M. brunneum* (BIPESCO5) resulted in a significant reduction in percentage disease incidence (Fig. 2a) as well as disease development (Fig. 2b) and severity (Fig. 2c). At the end of the experiments (24 dpi), all

F. culmorum-infected non-colonized plants (control) showed visible disease symptoms, whereas only 50 and 47% of *F. culmorum*-infected *B. bassiana*-colonized plants, and 60 and 50% of *F. culmorum*-infected *M. brunneum*-colonized plants developed visible CRR symptoms in Experiments 1 and 2, respectively (Fig. 2a). Furthermore, endophytic plant colonization with either fungal entomopathogen caused a significant delay in days to symptom development (Fig. 2b). Similarly, endophytic plant colonization with *B. bassiana* and *M. brunneum* significantly ($P < 0.0001$) reduced the severity of disease symptoms on shoots and roots of inoculated plants compared to control (Fig. 2c).

Notably, a significant negative effect of *F. culmorum* infection on almost all sampled growth parameters was observed at the end of the experiments (24 dpi). This negative effect was most pronounced in the absence of

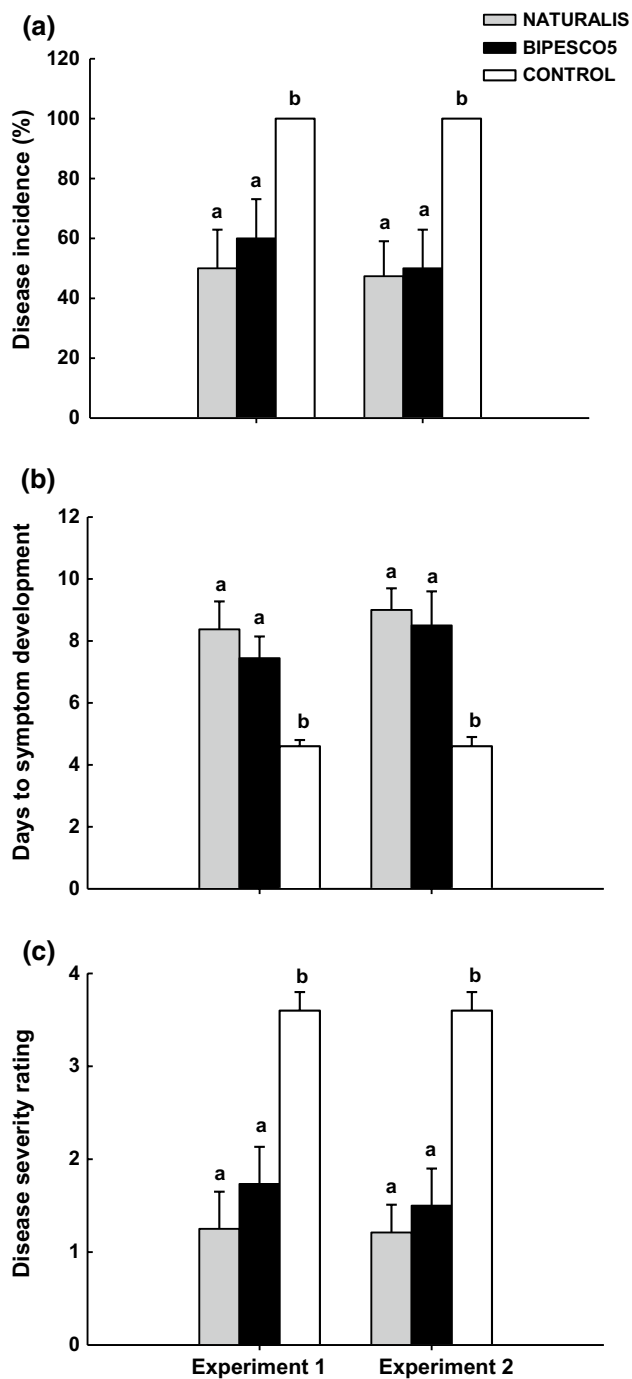


Fig. 2 Effect of endophytic colonization of wheat with fungal entomopathogens (*B. bassiana* NATURALIS and *M. brunneum* BIPESCO5) on **a** mean (±SE) percent CRR disease incidence, **b** mean (±SE) days to symptom development, and **c** mean (±SE) disease severity rating. Bars with different letters within the same experiment differ significantly at $P=0.05$ (Duncan's multiple range test after one-way ANOVA)

endophytic plant colonization with fungal entomopathogens (Table 3). On the other hand, a significant positive effect of endophytic colonization on plant growth was consistently

observed at 24 dpi. All sampled growth parameters, except fresh shoot weight in Experiment 1, of plants colonized with either *B. bassiana* or *M. brunneum* were significantly higher than those of non-colonized plants, irrespective of *Fusarium* infection (Table 3).

Discussion

This study demonstrates the ability of two fungal entomopathogens, *B. bassiana* and *M. brunneum*, to endophytically colonize roots and shoots of wheat following seed treatment. Plant colonization with both tested entomopathogens promoted several measured parameters of plant growth (e.g., seedling emergence, shoot height, root length, and fresh shoot and root weights). Previous studies have reported the colonization of wheat by *B. bassiana*, *Lecanicillium lecanii*, *Aspergillus parasiticus*, but not *M. brunneum* (Gurulingappa et al. 2010; Russo et al. 2015). Of these fungi, only *B. bassiana* was found to promote plant growth, whereas *L. lecanii* and *A. parasiticus* had no effect on the growth of wheat (Gurulingappa et al. 2010). The present study shows that, in addition to *B. bassiana*, *M. brunneum* can colonize wheat as an endophyte and enhance plant growth.

Soaking wheat seeds in the fungal inoculum of either *B. bassiana* or *M. brunneum* for 16 h resulted in a systemic colonization of root and shoot tissues of plants at 14 and 24 dpi. However, more root than shoot tissues were endophytically colonized with each inoculated fungal entomopathogen on both sampling dates. A similar systemic but differential colonization of plant tissues by fungal entomopathogens as endophytes, which are generally known to be adapted to particular conditions within a certain plant part or tissue (Petrini and Fisher 1987), was previously reported by other studies (Gurulingappa et al. 2010; Akutse et al. 2013; Jaber and Enkerli 2016, 2017; Jaber and Araj 2018). There was also a decline in plant colonization with *B. bassiana* and *M. brunneum* across both sampling dates. Such a decline in endophytic colonization over time has been demonstrated for entomopathogenic fungal strains, including those tested here, in other host plants as well (Posada et al. 2007; Gurulingappa et al. 2010; Jaber and Enkerli 2016, 2017; Jaber and Araj 2018), and could be due to plant tissue expansion or competition with other naturally occurring endophytes (Posada et al. 2007). Importantly, the extent of endophytic colonization by entomopathogenic fungi and the ability of these fungi to provide additional benefits such as improved plant growth have been shown to depend on the inoculation method, rate, and duration (Garcia et al. 2011; Jaber and Enkerli 2016, 2017; Jaber and Ownley 2018). Among the different inoculation methods used for establishing fungal entomopathogens as endophytes (e.g., Posada et al. 2007;

Gurulingappa et al. 2010; Russo et al. 2015), seed inoculation has been suggested as one of the most effective to initiate their role as plant growth promoters (Sasan and Bidochka 2012; Lopez and Sword 2015; Jaber and Enkerli 2016; Jaber and Ownley 2018). Results presented here support this suggestion by demonstrating a positive effect of endophytic *B. bassiana* and *M. brunneum*, inoculated into wheat via seed treatment, on percentage and speed of seedling emergence in addition to shoot height, root length, and fresh shoot and root weights. Such a suggestion has also been recently confirmed by Sun et al. (2018) who found that seed inoculation of radish with *Aspergillus oryzae* was effective in introducing the entomopathogenic fungus as an endophyte and enhancing plant height as well as dry weight for 7 weeks post-inoculation.

On the other hand, plant growth promotion by endophytic fungal entomopathogens does not seem to depend on nutrient availability. Many of the studies reporting a positive effect of endophytic colonization by entomopathogenic fungi on the growth of various host plants, including the present study, have been conducted under conditions of scarce nutrients, i.e., no fertilizers were added during the course of experiments (Sasan and Bidochka 2012; Lopez and Sword 2015; Jaber and Enkerli 2016, 2017; Jaber and Araj 2018; Sun et al. 2018). Similarly, Krell et al. (2018) have recently reported the ability of endophytic *M. brunneum* to improve dry plant biomass and leaf surface area under nutrient poor conditions. The study also shows that several measured physiological parameters (e.g., quantum yields of photosystem II, CO₂ assimilation, stomatal conductance, and water use efficiency) were significantly enhanced by *M. brunneum* endophytism, but only in nutrient poor conditions; whereas the positive effect of *M. brunneum* endophytism on the plant nutritional status (e.g., nitrogen and phosphate contents) was independent of soil fertility. The authors conclude that similar to mycorrhizal-plant beneficial interactions, which are negatively affected by mineral fertilization (Verbruggen et al. 2013), interactions with fungal entomopathogens as endophytes are more beneficial to plants when soil resources are limited (Krell et al. 2018). Contrary to these studies, Tall and Meyling (2018) reported that treatment of maize seeds with *B. bassiana* increased plant growth at high-nutrient conditions, whereas it reduced plant growth compared to control at low-nutrient conditions. The study indicates that the fungus may function as a sink when nutrients are limited. Indeed, to fit the definition of an endophyte (Wilson 1995), the fungus should colonize living plant tissues for all or part of its lifecycle without causing any apparent harm or symptomatic infection. Besides, unless there is a benefit of the fungal colonization to the plant, an endophyte becomes a parasite (or at least a commensal) that derives most or all of the host plant nutrients (Wilson 1995). Yet, Schulz and Boyle (2005) suggest that endophyte-plant

interactions occur along a continuum and range from mutualistic to parasitic. Important is the fact that the conditions under which known endophytic interactions can negatively impact the host plant is not well-understood and may depend on fungal genotype-plant genotype interaction in different environments (Jaber and Ownley 2018). In terms of potential mechanisms of plant growth promotion by endophytic fungal entomopathogens, previous studies indicate that these mechanisms might include an increased uptake of soil nutrients (Behie and Bidochka 2013), the transfer of insect-derived nitrogen back to plants (Behie and Bidochka 2014), and the upregulation of phytohormones (Khan et al. 2012a, b) or proteins related to photosynthesis and energy metabolism in inoculated plants (Gómez-Vidal et al. 2009).

In addition to their positive effect on plant growth when endophytic, *B. bassiana* and *M. brunneum* reduced symptoms of CRR caused by plant infection with *F. culmorum*. The disease incidence, development, and severity were significantly lower on plants colonized with the two tested fungal entomopathogens compared to control. Also following *F. culmorum* infection, the improvement in plant growth, as mediated by the endophytic colonization with either entomopathogen, was maintained in colonized *F. culmorum*-infected plants compared to non-colonized diseased plants. Similar results were obtained when applying *B. bassiana* as an endophyte to tomato and cotton seedlings against damping off (Ownley et al. 2008). Treatment of tomato and cotton seeds with *B. bassiana* 11–98 resulted in endophytic establishment of the fungus within seedlings and a subsequent suppression of the soil-borne pathogens, *R. solani* and *P. myriotylum* (Ownley et al. 2004, 2008). Percent stand counts and plant height of two tomato cultivars were also significantly higher in *B. bassiana*-colonized *R. solani*-infected plants compared to non-colonized *R. solani* control (Ownley et al. 2004). Furthermore, results obtained in the present study agree with others. For instance, bean plants colonized with *M. robertsii* then exposed to the root rot pathogen, *F. solani*, showed lower disease indices and healthier growth than plants not colonized with *M. robertsii* (Sasan and Bidochka 2013). Pre-treatment of cucumber roots with endophytic *Lecanicillium* spp. (formerly *Verticillium lecanii*) reduced another root rot pathogen, *P. ultimum* (Benhamou and Brodeur 2001). The study demonstrates that colonization of root cells with *Lecanicillium* restricted pathogen growth and triggered plant-induced resistance against the disease. Serfling et al. (2007) reported that wheat seedlings inoculated with the fungal endophyte *Piriformospora indica* then challenged 7 days later with *F. culmorum* had improved plant biomass and reduced disease severity. Similarly, wheat seedlings grown in soil inoculated with two strains of the Gram-negative bacterium *Pseudomonas* then infected with *F. culmorum* 10 days later showed significantly lower disease symptoms relative to seedlings only infected

with *F. culmorum* (Khan et al. 2006). The authors suggest that plant inoculation with the bacterial microbial agents resulted in the activation of systemic defence mechanisms against *F. culmorum*, as neither of the two tested bacterial strains showed direct antifungal activity against the fungal pathogen when tested in vitro. It is worth mentioning that the treatment regime (i.e., pre-treatment with microbial control agents prior to *F. culmorum* infection) used in this and other studies (e.g., Khan et al. 2006; Serfling et al. 2007) is not similar to a real field situation where treated seeds or transplants would immediately encounter the *Fusarium* pathogen (if present in soil) just after being placed in agricultural fields. However, in the present study, *F. culmorum* infection was done 14 days after planting seeds treated with either fungal entomopathogen to allow prior establishment of these entomopathogens as endophytes, otherwise the *Fusarium* pathogen would have possibly outcompeted the seed treatment during plant growth.

Plant disease suppression by endophytic fungal entomopathogens seems to involve multiple, but non-mutually exclusive, mechanisms (Ownley et al. 2010; Jaber and Ownley 2018), including those directly affecting plant pathogens (e.g., mycoparasitism, antibiosis, and competition), in addition to more complex ones that indirectly act against pathogens through mutual host plants (e.g., induction of systemic plant resistance or secondary metabolites and promotion of plant growth). Mycoparasitism has been shown to be one of the mechanisms employed by fungal entomopathogens against disease pathogens (Benhamou and Brodeur 2001; Ownley et al. 2008). This was observed for an endophytic strain of *B. bassiana* against *P. myriotylum* in parasitism assays (i.e., in vitro) (Ownley et al. 2008), and demonstrated for endophytic *Lecanicillium* against *P. ultimum* using both in vitro and in planta bioassays (Benhamou and Brodeur 2001). Mycoparasitism was, however, not observed when co-culturing either *B. bassiana* or *M. brunneum* and *F. culmorum* in dual plate assays as most recently reported by Jaber and Alananbeh (2018). On the other hand, the latter study shows that antibiosis (through production of fungal secondary metabolites) and competition for resources or niche are more likely to be the mechanisms through which both tested entomopathogenic fungal strains suppress *F. culmorum*. When simultaneously co-cultured in dual plate assays, *B. bassiana* and *M. brunneum* could be observed overgrowing *F. culmorum* with clear zones of inhibition occurring at the interface with the fungal pathogen. Such inhibition zones are known to result from secondary metabolites produced by either or both plated fungi (Yan et al. 2015). So far, several secondary metabolites, with antimicrobial and insecticidal activities, have been reported to be produced by *B. bassiana* and *Metarhizium* spp., including *M. brunneum*, in vitro as well as within endophytically colonized plants (i.e., in planta) (e.g., Ownley et al. 2010; Sasan and Bidochka 2013; Golo et al. 2014; Ríos-Moreno et al. 2016). Moreover,

systemic plant resistance has been demonstrated to be induced in cucumber plants endophytically colonized with *Lecanicillium* against *P. ultimum* (Benhamou and Brodeur 2001) and *P. fuliginea* (Hirano et al. 2008), in addition to cotton seedlings colonized with *B. bassiana* against *Xanthomonas* bacterial blight (Ownley et al. 2008). A similar induction of systemic resistance against *F. culmorum* in wheat plants colonized by *B. bassiana* or *M. brunneum* is also possible but remains to be determined. Alternatively, plant growth promotion by the two tested fungal entomopathogens may have played a role in *F. culmorum* suppression. Endophyte-mediated promotion of plant growth has been suggested to increase plant tolerance to a variety of abiotic and biotic stresses, including diseases caused by plant pathogens (reviewed in Kuldau and Bacon 2008; Gao et al. 2010; Jaber and Ownley 2018). In particular, increased plant growth due to endophytic colonization by either or both tested fungal strains has previously been shown to reduce ZYMV in squash (Jaber and Salem 2014) and Fusarium Crown and Root Rot (FCRR) caused by a complex of *Fusarium* spp. in sweet pepper (Jaber and Alananbeh 2018).

In conclusion, both tested fungal entomopathogens, *B. bassiana* and *M. brunneum*, can promote the growth of wheat following their endophytic establishment within plants through seed treatment. Results obtained here also show for the first time that, similar to endophytic *B. bassiana* which has already been reported as a disease antagonist by several previous studies, *M. brunneum* can suppress disease pathogens following plant colonization. These results confirm earlier reports on the multiple ecological roles that could further be played by entomopathogenic fungi, almost exclusively considered and used as insect pathogens. Bearing such additional roles in mind while developing these fungi as microbial agents would certainly improve the value of many commercially available mycoinsecticides (Vega et al. 2009; Lacey et al. 2015; Jaber and Ownley 2018).

Author contribution statement LRJ conceived, designed, and conducted the experiments as well as wrote the paper.

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